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METHOD OF MANUFACTURING LIGHT-EMITTING DEVICE, AND LIGHT-EMITTING DEVICE

Abstract

A light-emitting device includes: a plurality of light sources arrayed in a direction; and a bonding member bonding adjacent ones of the plurality of light sources to each other. Each of the light sources has an upper surface including a light-emitting portion, a lower surface opposite to the upper surface, and a lateral surface between the upper surface and the lower surface, wherein each of the light sources includes an external connection terminal at the lower surface. Lateral surfaces of the adjacent ones of the plurality of light sources are bonded via the bonding member such that the upper surfaces and the lower surfaces of the light sources are exposed from the bonding member and such that the bonding member is spaced apart from the external connection terminals. The upper surfaces of the light sources constitute a part of an upper surface of the light-emitting device.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a divisional application of U.S. patent application Ser. No. 17/551,389, filed on Dec. 15, 2021, which claims priority to Japanese Patent Application No. 2020-209994, filed on Dec. 18, 2020, and Japanese Patent Application No. 2021-146658, filed on Sep. 9, 2021. The disclosures of these applications are hereby incorporated by reference in their entirety.

BACKGROUND

[0002] The present disclosure relates to a method of manufacturing a light-emitting device and the light-emitting device.

[0003] A light source device has been disclosed that includes a collective substrate and a plurality of light-emitting portions. The collective substrate has a plurality of sections divided by partitions. The light-emitting portions are disposed in the sections and have different emission colors (see Japanese Patent Publication No. 2013-183042).

SUMMARY

[0004] According to certain embodiments of the present disclosure, an efficient method of manufacturing a light-emitting device including a plurality of light-emitting portions, and the light-emitting device can be provided.

[0005] A method of manufacturing a light-emitting device according to an embodiment of the present disclosure includes: a providing step in which a plurality of light sources are provided, each of the light sources having an upper surface including a light-emitting portion, a lower surface opposite to the upper surface and provided with an external connection terminal, and lateral surfaces between the upper surface and the lower surface, the plurality of light sources being ranked in terms of at least one of luminous flux and chromaticity; an extracting step in which a plurality of light sources in a desired rank are extracted from the plurality of light sources; and a bonding step in which the lateral surfaces of adjacent ones of the plurality of extracted light sources are bonded via a bonding member such that the upper surfaces and the lower surfaces of the light sources are exposed from the bonding member and such that the bonding member is spaced apart from the external connection terminals.

[0006] A light-emitting device according to an embodiment of the present disclosure includes a plurality of light sources; and a bonding member bonding the plurality of light sources. Each of the light sources having an upper surface including a light-emitting portion, a lower surface opposite to the upper surface and provided with an external connection terminal, and a lateral surface between the upper surface and the lower surface. Lateral surfaces of adjacent ones of the plurality of light sources are bonded via a bonding member such that the upper surfaces and the lower surfaces are exposed from the bonding member and such that the bonding member are spaced apart from the external connection terminals.

[0007] The present disclosure allows for providing an efficient method of manufacturing a light-emitting device, and the light-emitting device including a plurality of light-emitting portions.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1A is a schematic perspective view illustrating a configuration of a light-emitting device according to a first embodiment.

[0009] FIG. 1B is a schematic plan view illustrating the configuration of the light-emitting device according to the first embodiment.

[0010] FIG. 1C is a schematic cross-sectional view taken along line IC-IC of FIG. 1B.

[0011] FIG. 1D is a schematic cross-sectional view taken along line ID-ID of FIG. 1B.

[0012] FIG. 1E is a schematic bottom view illustrating the configuration of the light-emitting device according to the first embodiment.

[0013] FIG. 1F schematically shows an example of the light-emitting device according to the first embodiment.

[0014] FIG. 2 is a flowchart of a method of manufacturing the light-emitting device according to the first embodiment.

[0015] FIG. 3A is a schematic cross-sectional view illustrating the method of manufacturing the light-emitting device according to the first embodiment.

[0016] FIG. 3B is a schematic cross-sectional view illustrating the method of manufacturing the light-emitting device according to the first embodiment.

[0017] FIG. 3C is a schematic cross-sectional view illustrating the method of manufacturing the light-emitting device according to the first embodiment.

[0018] FIG. 3D is a schematic cross-sectional view illustrating the method of manufacturing the light-emitting device according to the first embodiment.

[0019] FIG. 3E is a schematic cross-sectional view illustrating the method of manufacturing the light-emitting device according to the first embodiment.

[0020] FIG. 3F is a schematic cross-sectional view illustrating the method of manufacturing the light-emitting device according to the first embodiment.

[0021] FIG. 3G is a schematic cross-sectional view illustrating the method of manufacturing the light-emitting device according to the first embodiment.

[0022] FIG. 4A is a schematic plan view of a light-emitting device according to a second embodiment.

[0023] FIG. 4B is a schematic cross-sectional view taken along line IVB-IVB of FIG. 4A.

[0024] FIG. 5 is the 1931 CIE chromaticity diagram illustrating the chromaticity of the light-emitting device according to the second embodiment.

[0025] FIG. 6A schematically shows a light-emission state of the light-emitting device according to the second embodiment.

[0026] FIG. 6B schematically shows a light-emission state of the light-emitting device according to the second embodiment.

[0027] FIG. 6C schematically shows a light-emission state of the light-emitting device according to the second embodiment.

DETAILED DESCRIPTION

[0028] Certain embodiments are described below with reference to the accompanying drawings.

The embodiments described below are intended to exemplify methods of manufacturing light-emitting devices and the light-emitting devices to give a concrete form to the technical idea of the present invention. However, the present invention is not limited to the embodiments described below. Unless specifically stated otherwise, descriptions of the sizes, materials, shapes, and relative

positions of constituent components in the embodiments described below are not intended to limit the scope of the present invention to those descriptions, but are rather only examples. Sizes or positional relationships of components illustrated in the drawings may be exaggerated in order to clarify the descriptions. End views showing only cut-surfaces may be used as the cross-sectional views. The number of light-emitting elements shown in each drawing is selected as an example to facilitate understanding of the configuration. Unless specifically stated otherwise, the term “cover” used in embodiments includes not only being in direct contact but also includes indirect covering, such as covering via another member.

First Embodiment

[0029] FIG. 1A is a schematic perspective view illustrating the configuration of a light-emitting device according to a first embodiment. FIG. 1B is a schematic plan view illustrating the configuration of the light-emitting device according to the first embodiment. FIG. 1C is a schematic cross-sectional view taken along the line IC-IC of FIG. 1B. FIG. 1D is a schematic cross-sectional view taken along the line ID-ID of FIG. 1B. FIG. 1E is a schematic bottom view illustrating the configuration of the light-emitting device. FIG. 1F schematically shows a combination of ranked light sources.

Light-Emitting Device

[0030] A light-emitting device **200** includes a plurality of light sources **100**, and bonding members **60**, each bonding corresponding ones of the light sources **100**. Each of the light sources **100** has an upper surface **101** including a light-emitting portion **110**, a lower surface **102** opposite to the upper surface **101** and provided with external connection terminals **3**, and lateral surfaces **103** between the upper surface **101** and the lower surface **102**. For example, the light source **100** has a substantially rectangular parallelepiped shape. Lateral surfaces **103** of adjacent light sources **100** are bonded with the bonding members **60** disposed therebetween such that the upper surfaces **101** and the lower surfaces **102** of the light sources **100** are exposed from the bonding members **60** and such that the bonding members **60** are spaced apart from the external connection terminals **3**, which constitutes the single light-emitting device **200**. In this example, lateral surfaces of three light sources **100** having the same emission color are bonded together such that the light-emitting portions **110** are arranged in one row and three columns to constitute the light-emitting device **200**. [0031] The light sources **100** of the light-emitting device **200** may have the same emission color or different emission colors. For example, when using a combination of light sources having the same emission color, unevenness in emission color of the light-emitting portions of the light-emitting device can be reduced by selecting and combining light sources in closer chromaticity ranks. By combining different emission colors, a multicolor light-emitting device can be obtained.

Light Source

[0032] The light source **100** has a substantially rectangular parallelepiped shape and includes the light-emitting portion in the upper surface and the external connection terminals **3** on the lower surface for connection to an external device. The light source **100** includes a substrate **10**, a light-emitting element disposed on the substrate **10**, a light-transmissive member **30** covering the light-emitting element, and a covering member **50** covering the lateral surfaces of the light-transmissive member, the upper surface of the light-transmissive member being exposed from the covering member **50**. The upper surface of the light-transmissive member **30** exposed from the covering member **50** serves as the light-emitting portion of the light source **100**, and light from the light-emitting element is emitted to the outside through the upper surface of the light-transmissive member **30**. The covering member **50** constitutes a portion of the upper surface and at least a portion of outer lateral surfaces of the light source.

Substrate

[0033] The substrate **10** is a member on which a light-emitting element **20** is to be disposed. For example, the substrate **10** has a substantially rectangular shape in a plan view. The substrate **10** may have a flat plate shape or may define a recess in the upper surface to accommodate the light-

emitting element **20**.

[0034] The substrate **10** includes a plurality of wirings that function as electrodes for the light source and as an insulating base material. Examples of the substrate **10** include a ceramic substrate including a wiring layer and a resin package in which lead electrodes and resin are integrally molded.

[0035] For the base material, a thermoplastic resin such as polyphthalamide (PPA), poly(phenylene sulfide) (PPS), or a liquid crystal polymer or a thermosetting resin such as an epoxy resin, a silicone resin, a modified epoxy resin, a modified silicone resin, a urethane resin, or a phenolic resin can be used. In the case in which a ceramic is used for the base material, aluminum oxide, aluminum nitride, mullite, silicon carbide, or silicon nitride is preferably used.

[0036] Examples of the wirings include metals such as Fe, Cu, Ni, Al, Ag, Au, Pt, Ti, W, and Pd and alloys containing at least one of these metals. The wirings may include a plating layer on surfaces thereof. For example, Au, Ag, Cu, Pt, or an alloy containing one of these metals can be used for the plating layer.

[0037] The wirings include upper-surface wirings **2**, the external connection terminals **3**, and inner-layer wirings. The upper-surface wirings **2** are disposed on the upper surface of the substrate and connected to the light-emitting element **20**. The external connection terminals **3** are disposed on the lower surface opposite to the upper surface and electrically connected to an external power source. Each of the inner-layer wirings electrically connects a corresponding one of the upper-surface wirings **2** to a corresponding one of the external connection terminals **3**. For example, each of the inner-layer wirings includes vias **4** extending through the base material.

Light-Emitting Element and Protective Element

[0038] For the light-emitting element **20**, a light-emitting diode is preferably used. Any appropriate shape, size, and the like of the light-emitting element **20** can be selected. A light-emitting element **20** having any appropriate wavelength can be selected. For example, a light-emitting element containing a nitride semiconductor (In.sub.XAl.sub.YGa.sub.1-X-YN, where $0 \leq X$, $0 \leq Y$, $X+Y \leq 1$), ZnSe, or GaP can be used for a blue or green light-emitting element **20**. For a red light-emitting element **20**, a nitride semiconductor element, an element including GaAlAs, AlInGaP, or the like can be used. When a light-emitting element is used in combination with a wavelength conversion member, it is preferable to use a light-emitting element containing a nitride semiconductor adapted to emit short-wavelength light for efficient excitation of the wavelength conversion member.

[0039] For example, the light-emitting element **20** includes a semiconductor layer on a light-transmissive supporting substrate such as a sapphire growth substrate. The light-emitting element **20** does not necessarily include the supporting substrate.

[0040] The light-emitting element **20** includes positive and negative element electrodes. The positive and negative element electrodes may be disposed on the same surface side or different surface sides. A light-emitting element **20** with a desired electrode configuration can be appropriately selected according to the form of the substrate used for the light source and the like. The light-emitting element **20** is electrically connected to the upper-surface wirings of the substrate by respective electroconductive connecting members **8**. Examples of the electroconductive connecting members **8** include eutectic solder, electroconductive paste, sintered metal, bumps, and wires.

[0041] For a protective element **25**, for example, a Zener diode can be used. For example, the protective element **25** having positive and negative electrodes on a single surface is mounted on the upper-surface wirings of the substrate **10** via electroconductive connecting members. The light source **100** does not necessarily include the protective element **25**.

Light-Transmissive Member

[0042] The light-transmissive member **30** covers the light-emitting element **20** and transmits light emitted from the light-emitting element **20** to the outside. In the upper surface of the light source **100**, the upper surface of the light-transmissive member **30** is exposed from the covering member

50 to serve as the light-emitting portion of the light source. For example, the light-transmissive member **30** transmits 60% or more, preferably 70% or more, of light emitted from the light-emitting element **20** and/or light emitted from the light-emitting element **20** and subjected to wavelength conversion (such as light within the wavelength range of 320 nm to 850 nm). For example, such a member is made of an inorganic material such as glass, ceramic, or sapphire or an organic material such as a resin or a hybrid resin containing one or more of a silicone resin, a modified silicone resin, an epoxy resin, a modified epoxy resin, an acrylic resin, a phenolic resin, and a fluorocarbon resin.

[0043] The light-transmissive member **30** may contain a phosphor adapted to convert a wavelength of at least a portion of light incident on the light-transmissive member **30**. Examples of the light-transmissive member **30** containing a phosphor include a mixture of the above-described material and phosphor powder, and a sintered body of a phosphor. For the light-transmissive member **30**, a molded body made of resin, glass, or ceramic with a phosphor layer such as a resin layer containing a phosphor or a glass layer containing a phosphor on a surface of the molded body may be employed. The light-transmissive member **30** may contain a filler such as a diffusing material according to the purpose. When the light-transmissive member **30** contains a filler such as a diffusing material, the materials described above with the filled contained therein, or a molded body made of resin, glass, or ceramic and provided with a diffusing material layer, such as a resin layer containing a filler or a glass layer containing a filler, disposed on a surface of the molded body may be used for the light-transmissive member **30**.

[0044] For the phosphor, a substance known in the art can be used. Examples of a phosphor adapted to emit green light include yttrium-aluminum-garnet based phosphors (such as $\text{Y}_{0.3}\text{Al}_{0.7}\text{O}_{12}\text{Ce}$), lutetium-aluminum-garnet based phosphors (such as $\text{Lu}_{0.3}\text{Al}_{0.7}\text{O}_{12}\text{Ce}$), terbium-aluminum-garnet based phosphors (such as $\text{Tb}_{0.3}\text{Al}_{0.7}\text{O}_{12}\text{Ce}$), silicate based phosphors (such as $(\text{Ba},\text{Sr})_2\text{SiO}_4\text{Eu}$), chlorosilicate based phosphors (such as $\text{Ca}_8\text{Mg}(\text{SiO}_4)_4\text{Cl}_2\text{Eu}$), β -SiAlON based phosphors (such as $\text{Si}_{6-z}\text{Al}_z\text{O}_z\text{N}_{8-z}\text{Eu}$ ($0 < z \leq 4.2$)), and SGS based phosphors (such as $\text{SrGa}_2\text{S}_4\text{Eu}$).

[0045] Examples of a phosphor that emits yellow light include α -SiAlON phosphors (such as $\text{M}_z(\text{Si},\text{Al})_{12}(\text{O},\text{N})_{16}$ (where $0 < z \leq 2$, and M is Li, Mg, Ca, Y, or a lanthanoid element except for La and Ce)). The phosphors adapted to emit green light described above also include phosphors adapted to emit yellow light. Also, yellow light can be obtained by, for example, substituting a portion of Y in an yttrium-aluminum-garnet phosphor with Gd to shift the peak emission wavelength to a longer wavelength. These phosphors include fluorescent substances adapted to emit orange light. Examples of a phosphor adapted to emit red light include nitrogen-containing calcium aluminosilicate (CASN or SCASN) based phosphors (such as $(\text{Sr},\text{Ca})\text{AlSiN}_3\text{Eu}$) and BSESN based phosphors (such as $(\text{Ba},\text{Sr},\text{Ca})_2\text{Si}_5\text{N}_8\text{Eu}$). Examples of the phosphor adapted to emit red light also include manganese-activated fluoride phosphors (phosphors represented by a general formula (I) $\text{A}_2[\text{M}_{1-a}\text{Mn}_a\text{F}_6]$ (in the general formula (I), A is at least one selected from the group consisting of K, Li, Na, Rb, Cs, and NH_4 , M is at least one element selected from the group consisting of the group IV elements and the group XIV elements, and “a” satisfies $0 < a < 0.2$)). Examples of the manganese-activated fluoride phosphors include manganese-activated potassium fluorosilicate phosphors (such as $\text{K}_2\text{SiF}_6\text{Mn}$), KSAF-based phosphors (for example, $\text{K}_{0.99}\text{Si}_{0.99}\text{Al}_{0.01}\text{F}_{5.99}\text{Mn}$), and MGF-based phosphors (for example, $3.5\text{MgO}, 0.5\text{MgF}_2, \text{GeO}_2\text{Mn}$). The KSAF-based phosphor can have a composition represented by formula (I) shown below.

$\text{M}_2[\text{Si}_p\text{Al}_q\text{MN}_r\text{F}_s]$ (I)

[0046] In the formula (I), M represents an alkaline metal and may contain at least K. Mn may be a

tetravalent Mn ion. p, q, r, and s may satisfy $0.9 < p+q+r \leq 1.1$, $0 < q \leq 0.1$, $0 < r \leq 0.2$, and $5.9 \leq s \leq 6.1$. p, q, r, and s preferably satisfy $0.95 \leq p+q+r \leq 1.05$ or $0.97 \leq p+q+r \leq 1.03$, $0 < q \leq 0.03$, $0.002 \leq q \leq 0.02$, or $0.003 \leq q \leq 0.015$, $0.005 \leq r \leq 0.15$, $0.01 \leq r \leq 0.12$, or $0.015 \leq r \leq 0.1$, $5.92 \leq s \leq 6.05$ or $5.95 \leq s \leq 6.025$. For example, a composition represented by

K.sub.2[Si.sub.0.946Al.sub.0.005Mn.sub.0.049F.sub.5.995], a composition represented by K.sub.2[Si.sub.0.942Al.sub.0.008Mn.sub.0.050F.sub.5.992], or a composition represented by K.sub.2[Si.sub.0.939Al.sub.0.014Mn.sub.0.047F.sub.5.986] can be employed. With such a KSAF-based phosphor, a red light emission having a high luminance and a narrow half width of the emission peak can be obtained.

[0047] For the diffusing material, a substance known in the art can be used. For example, barium titanate, titanium oxide, aluminum oxide, or silicon oxide can be used.

[0048] When a resin is used as a binder for the phosphor or the diffusing material, examples of the resin include the resin used for the light-transmissive member **30** described above.

Light-Guiding Member

[0049] The light source may include a light-guiding member **40** between the light-transmissive member **30** and the light-emitting element **20**.

[0050] For example, the light-guiding member **40** bonds the light-emitting element **20** to the plate-shaped or sheet-shaped light-transmissive member **30**. The light-guiding member **40** is configured to guide light from the light-emitting element **20** to the light-transmissive member **30**. Accordingly, a light-transmissive material is preferably used for the light-guiding member **40**. The light-guiding member **40** may also be located on the lateral surfaces of the light-emitting element **20**.

[0051] For the light-guiding member **40**, for example, a light-transmissive resin can be used. Examples of the light-transmissive resin include resins described above as examples of a material to be used for the light-transmissive member **30**, and the like. The light-guiding member **40** may contain the phosphor and/or the diffusing material described above.

Covering Member

[0052] The covering member **50** covers the lateral surfaces of the light-transmissive member **30** such that the upper surface of the light-transmissive member **30** is exposed. The covering member **50** is disposed on the flat plate-shaped substrate **10** and covers the lateral surfaces of the light-emitting element **20** and the lateral surfaces of the light-transmissive member **30**. In the case in which the lateral surfaces of the light-emitting element **20** are covered with the light-guiding member **40** and the light-transmissive member **30**, the covering member covers the lateral surfaces of the light-emitting element **20** with the light-guiding member **40** and the light-transmissive member **30** located therebetween.

[0053] For the covering member **50**, for example, a resin can be used. Examples of the resin used for the covering member **50** include the resin used for the light-transmissive member **30** described above.

[0054] The covering member **50** may contain a filler such as a light-reflective substance in the resin described above. This allows the covering member **50** to serve as a light-reflective member for reflecting light emitted from the light-emitting element and efficiently emitting the light upward.

[0055] Examples of the light-reflective substance include titanium oxide, silicon oxide, aluminum oxide, zirconium oxide, magnesium oxide, potassium titanate, zinc oxide, silicon nitride, and boron nitride. Among these materials, titanium oxide, which has a relatively high refractive index, is preferably used in view of reflection of light. For example, the proportion of the light-reflective substance can be 20 wt % to 80 wt % or less relative to the weight of the resin.

[0056] The covering member can be formed integrally with the substrate. For example, when using a substrate defining a recess in the upper surface of the substrate, the recess accommodating the light-emitting element **20**, the covering member may be a lateral wall defining the recess and formed integrally with the base material.

[0057] A portion of the covering member **50** covering the lateral surfaces of the light-transmissive

member **30** preferably has a width in an array direction of the light sources (that is, the distance between a lateral surface of the light-transmissive member **30** and an outer lateral surface of the light source) of 30 μm or greater and 300 μm or less, more preferably 50 μm or greater and 200 μm or less, in view of inhibition of leakage of light in a lateral direction and miniaturization of the light source.

Bonding Member

[0058] The bonding members **60** bond a plurality of light sources **100**. The bonding members **60** are disposed between lateral surfaces of adjacent light sources **100** facing each other. Each of the bonding members **60** bonds corresponding lateral surfaces **103** of adjacent light sources **100** such that the upper surfaces **101** and the lower surfaces **102** of the light sources **100** are exposed and such that the bonding members **60** are spaced apart from the external connection terminals **3**. That is, in the light-emitting device **200**, lateral surfaces of adjacent ones of a plurality of light sources **100** are bonded with the bonding members disposed therebetween such that the upper surfaces of light sources **100** with the light-emitting portions and the lower surfaces of light sources **100** with the external connection terminals of the light sources are exposed from the bonding members.

[0059] The bonding members **60** are disposed between adjacent light sources. In order to reduce interference of light between adjacent light sources, a member having a light transmittance lower than the light transmittance of the covering member **50** is preferably used for the bonding member **60**. In this case, the bonding member **60** more preferably covers substantially the entirety of lateral surfaces facing each other in adjacent light sources.

[0060] For the bonding member **60**, for example, a resin containing a light-shielding filler can be used. Examples of the resin include resins described above as examples of the resin to be used for the light-transmissive member **30**. Examples of the light-shielding filler include fillers such as light-absorbing substances and light-reflective substances. Examples of the light-reflective substance include light-reflective substances described above as examples of the light-reflective substance to be used for the covering member **50**. Examples of the light-absorbing substance include black pigments such as carbon black and graphite. A pigment or a dye of a color other than black may also be added. For example, a gray resin in which black and white pigments have been added may be used. Among these materials, a black resin or a gray resin is preferably used. When the bonding member **60** made of a black resin or a gray resin is disposed between the light sources **100**, interference of light from a light source **100** with an adjacent light source **100** can be reduced. This structure allows for obtaining a light-emitting device with good “visibility,” which refers to good contrast between a light-emitting region and a non-light-emitting region when the light sources are individually turned on.

[0061] The bonding member **60** is preferably disposed between the light sources such that the upper surface of the bonding member **60** is in the same plane with the upper surface of the light source, considering creeping up of the bonding member **60** onto the upper surface of the light source. The upper surface of the bonding member **60** may be slightly recessed considering thermal expansion of the bonding member **60**.

[0062] The bonding member **60** is preferably disposed between the light sources such that the lower surface of the bonding member is in the same plane with the lower surface of the light source, considering creeping up of the bonding member onto the external connection terminals. The lower surface of the bonding member **60** may be slightly recessed considering thermal expansion of the resin.

[0063] The recessed surface can be formed using a sink mark due to curing of the resin (shrinkage of the resin).

[0064] A portion of the bonding member **60** between the light sources preferably has width (that is, the distance between adjacent light sources) of 1 μm or greater and 200 μm or less, greater preferably 5 μm or greater and 50 μm or less, in view of reduction of interference of light among the light sources, reduction of dark lines between the light sources when the adjacent light sources

are turned on at the same time, and ease of mixing of emission colors when light sources having different emission colors are combined.

[0065] The light source **100** can include a light-emitting portion having a desired emission color by adjusting the emission wavelength of the light-emitting element and the combination and blending ratio of wavelength conversion members contained in the light-transmissive member. An example of a red light source is a light source including a blue light-emitting element **20** and a light-transmissive member **30** containing a red phosphor. An example of a white light-emitting portion is a light-emitting portion including a blue light-emitting element **20** and a light-transmissive member **30** containing a yellow phosphor. An example of a green light-emitting portion is a light-emitting portion including a green light-emitting element **20** and a light-transmissive member **30** containing a diffusing material. An example of a green light-emitting portion is a light-emitting portion including a blue light-emitting element **20** and a light-transmissive member **30** containing a green phosphor. An example of a blue light-emitting portion is a light-emitting portion including a blue light-emitting element **20** and a light-transmissive member **30** containing a diffusing material.

[0066] As described below, light sources in a desired rank are extracted from light sources ranked in terms of at least one of luminous flux and chromaticity and combined, which constitutes the light sources **100** of the light-emitting device **200**.

[0067] For example, the light sources **100** of the light-emitting device **200** are a combination of a plurality of light sources **100** extracted from a group of light sources having the same emission color, such as light sources in the same chromaticity rank. This can reduce unevenness in emission color of the light-emitting device **200**.

[0068] In this case, light sources in the same chromaticity rank and different luminous flux ranks may be combined. For example, a light source **100** at a center of the light emitting device **200** can be in a higher luminous flux rank than light sources **100** disposed at peripheral sides of the light emitting device **200**. As shown in FIG. 1F, for example, the single light source **100** disposed at the center is preferably in a higher rank than the light sources **100** disposed at the peripheral sides, and the light sources disposed at the peripheral sides are preferably in an equal luminous flux rank. For example, the luminous flux of the light source disposed at the center is 400 to 425 lumens, and the luminous flux of the light sources disposed at the peripheral sides is 375 to 400 lumens. Such light-emitting characteristics allow for illuminating more brightly the central region of an irradiation region when using for a vehicle lamp such as a headlight, which is preferable. The luminous flux rank of light sources to be extracted can be selected considering the optical characteristics of a lens to be combined with the light-emitting device.

Method of Manufacturing Light-Emitting Device

[0069] A method of manufacturing the light-emitting device will be described.

[0070] FIG. 2 is a flowchart of the method of manufacturing the light-emitting device according to the first embodiment. FIG. 3A to FIG. 3G are schematic cross-sectional views illustrating the method of manufacturing the light-emitting device according to the first embodiment.

[0071] The method of manufacturing the light-emitting device **200** includes a providing step **S104**, an extracting step **S105**, and a bonding step **S106**. In the providing step **S104**, a plurality of light sources **100** are provided. Each of the light sources **100** has the upper surface **101** including the light-emitting portion **110**, the lower surface **102** opposite to the upper surface **101**, and the lateral surfaces **103** between the upper surface **101** and the lower surface **102** and is ranked in terms of at least one of luminous flux and chromaticity. In the extracting step **S105**, a plurality of light sources **100** in a desired rank are extracted from the light sources **100**. In the bonding step **S106**, the lateral surfaces **103** of the extracted light sources **100** are bonded via the bonding members **60** such that the upper surfaces **101** and the lower surfaces **102** of the light sources **100** are exposed from the bonding members **60**.

[0072] The method of manufacturing the light-emitting device **200** may also include, before the providing step **S104**, a collective body providing step **S101**, a singulation step **S102**, and a

classification step **S103**. In the collective body providing step **S101**, a collective body **150** of light sources including a plurality of light-emitting portions **110** in the upper surface is provided. In the singulation step **S102**, the collective body **150** is divided between adjacent ones of the light-emitting portions **110** into a plurality of light sources **100**. In the classification step **S103**, luminous flux and chromaticity of the light sources **100** after the singulation step **S102** are measured, and the light sources **100** are ranked on the basis of the at least one of luminous flux and chromaticity measured.

[0073] The materials, arrangement, and the like of members are as described for the light-emitting device **200** described above, and detailed description thereof may be omitted when appropriate.

Collective Body Providing Step

[0074] In the collective body providing step **S101**, the collective body **150** including a plurality of light-emitting portions **110** on the upper surface is provided as the collective body **150** of light sources. The collective body **150** includes a collective substrate **11** that is a collective body of substrates **10** of light sources **100**, a plurality of light-emitting elements **20** disposed on the collective substrate **11**, light-transmissive members each covering a respective one of the light-emitting elements, and a covering member covering lateral surfaces of the light-transmissive members such that the upper surfaces of the light-transmissive members are exposed.

[0075] In the step **S101**, the collective substrate **11** is provided including a plurality of disposition regions **12** on the upper surface in which the light-emitting elements **20** are to be disposed. At least one light-emitting element **20** is disposed in each of the disposition regions **12**. The collective substrate **11** includes wirings on each of the upper surface and the lower surface opposite to the upper surface. A protective element **25** may be disposed in each of the disposition regions **12**. FIG. 4A schematically shows a collective substrate **11** including eight disposition regions **12** aligned in a lateral direction for convenience, but the number of disposition regions **12** can be appropriately adjusted.

[0076] Subsequently, the light-transmissive members **30** are disposed to cover respective light-emitting elements **20** each disposed in a respective one of the disposition regions **12**. For example, the light-transmissive member **30** can be formed by potting, printing, or spraying. Alternatively, the light-transmissive members **30** that have been formed into plates or sheets may be provided and disposed on the light-emitting elements **20**. In this example, plate-shaped light-transmissive members **30** are disposed on the light-emitting elements **20** provided with light-guiding members on the upper surfaces of the light-emitting elements **20**, and each of the light-emitting elements **20** and a respective one of the light-transmissive members **30** are bonded together with a respective one of the light-guiding members **40** disposed therebetween.

[0077] Subsequently, a resin containing a light-reflective member is disposed on the collective substrate **11** as the covering members **50** covering the lateral surfaces of the light-transmissive members **30**. The resin is disposed on the collective substrate **11** such that the upper surfaces of the light-transmissive members **30** are exposed from the resin. That is, the upper surfaces of the light-transmissive members **30** are exposed from the covering members **50**, and serve as the light-emitting portions. The collective body **150** of light sources including a plurality of light-emitting portions **110** on the upper surface can thus be formed. The covering members **50** cover the lateral surfaces of the light-emitting elements **20**, the lateral surfaces of the light-guiding members **40**, the lateral surfaces of the light-transmissive members **30** and the upper surface of the collective substrate **11**. The covering members **50** may be disposed in the gaps between the light-emitting elements **20** and the collective substrate **11** to serve as underfill. For example, the covering members **50** can be formed by transfer molding, injection molding, compression molding, potting, printing, or spraying. Alternatively, a collective substrate defining a plurality of recesses in the upper surface thereof may be provided as the collective substrate **11**, one or more light-emitting elements may be disposed at the bottom of each recess, and the light-transmissive member **30** covering the light-emitting element may be disposed in each of the plurality of recesses.

Singulation Step

[0078] In the singulation step **S102**, the collective body **150** of light sources is divided between adjacent ones of the light-emitting portions **110** into a plurality of light sources **100**. In the singulation step **S102**, the covering member **50** and the collective substrate **11** are divided in the thickness direction between adjacent light-emitting portions. For example, the division can be performed by cutting with a blade. This allows for obtaining a plurality of light sources **100** each having the upper surface **101** including the light-emitting portion **110**, the lower surface **102** opposite to the upper surface **101**, and the lateral surfaces **103** between the upper surface **101** and the lower surface **102**.

Classification Step

[0079] In the classification step **S103**, the luminous flux and/or chromaticity of each light source **100** after the singulation step **S102** is measured, and each light source **100** is ranked on the basis of the measured value. Alternatively, light sources provided by purchasing or the like may be ranked in the classification step **S103**.

[0080] Classification in terms of luminous flux can be performed on the basis of measured luminous fluxes (in units of lumens) into three ranks such as a high luminous flux rank assigned to light sources **100H**, a middle luminous flux rank assigned to light sources **100M**, and a low luminous flux rank assigned to light sources **100L**. The classification may be performed into two ranks or four or more ranks according to desired optical characteristics of the light-emitting device **200** and the arrangement of the light sources **100**.

[0081] Classification in terms of chromaticity can be performed on the basis of any appropriate chromaticity ranges (such as a range defined by a quadrangle connecting four chromaticity coordinates) set on the basis of XY chromaticity coordinates of the CIE 1931 chromaticity diagram.

Providing Step

[0082] In the providing step **S104**, a plurality of light sources each of which has an upper surface including a light-emitting portion, a lower surface opposite to the upper surface, and lateral surfaces between the upper surface and the lower surface and is ranked in terms of at least one of luminous flux and chromaticity are provided. The light sources may be provided by purchasing light sources ranked in terms of at least one of luminous flux and chromaticity, or the light sources **100** ranked in the classification step **S103** may be provided. For example, in the providing step **S104**, white light sources in the same chromaticity class (rank) are provided that include the light sources **100H** in a high luminous flux rank, the light sources **100L** in a lower luminous flux rank than the light source **100H**, and the light sources **100M** in a rank between the rank of the light sources **100H** and the rank of the light sources **100L**.

Extracting Step

[0083] In the extracting step **S105**, a plurality of light sources **100** in a desired rank are extracted from the light sources **100** provided in the step **S104**. Light sources in an appropriate rank are extracted such that the light-emitting device **200** has desired optical properties. For example, when producing a light-emitting device **200** in which a plurality of white light sources **100** are arranged, two ranks of light sources **100** are selected; that is, a light source **100H** in the high luminous flux rank and light sources **100M** in the middle luminous flux rank are extracted. The extracted light sources **100** in two ranks are arranged in a lateral direction such that the light source **100H** in the high luminous flux rank is located between the light sources **100M** in the middle luminous flux rank.

[0084] In the example shown in FIG. 1F, the light sources **100** are arranged in one row and three columns, in which the light source in the high luminous flux rank is disposed at the center. Alternatively, a plurality of light sources **100** may be arranged in a matrix other than the one-by-three matrix.

Bonding Step

[0085] In the bonding step **S106**, the lateral surfaces **103** are bonded together with the bonding

members **60** disposed therebetween such that the upper surfaces **101** and the lower surfaces **102** of the light sources **100** extracted and arranged are exposed from the bonding members **60**.

[0086] In the bonding step **S106**, the light sources **100** extracted in the extracting step **S105** are disposed on a supporting member, and uncured bonding materials are disposed between the light sources **100** by, for example, potting or spraying, and are cured to form the bonding members **60**. The bonding members **60** bond corresponding lateral surfaces of the light sources **100**, such as cut surfaces formed in the singulation step **S102**. Bonding the cut surfaces, which tend to be rough, allows for improving adhesion to the bonding members. The bonding members **60** are spaced apart from the external connection terminals **3** so as not to cover the external connection terminals **3** on the lower surfaces of the light sources **100**. In the bonding step **S106**, the external connection terminals **3** may be covered with the bonding members **60**, and excessive portions of the bonding members **60** covering the external connection terminals **3** may be removed.

[0087] Examples of the supporting member include sheet members constituted of heat-resistant resin sheets or UV-curable sheets. When a resin sheet is used for the supporting member, the bonding step **S106** may include a step of removing the sheet member from the lower surfaces of the light sources **100** bonded together with the bonding members **60**.

[0088] In the case in which the collective substrate is cut with, for example, a blade in the singulation step described above, a width (that is, the distance between the lateral surfaces of adjacent light sources **100** facing each other) of the bonding member disposed between lateral surfaces of adjacent ones of the light sources **100** facing each other can be smaller than a cut width of the blade, so that the light sources **100** can be disposed more closely to each other. This allows for miniaturizing the light-emitting device to be obtained.

Second Embodiment

[0089] Next, a light-emitting device according to a second embodiment will be described with reference to FIG. 4A, FIG. 4B, FIG. 5, FIG. 6A, and FIG. 6B. FIG. 4A is a schematic plan view of the light-emitting device according to the second embodiment, and FIG. 4B is a schematic cross-sectional view taken along line IVB-IVB of FIG. 4A.

[0090] A light-emitting device **200T** includes a plurality of light sources **100T1** and **100T2** (two in this case) and a bonding member **60T** bonding the light sources **100T1** and **100T2**. Each of the light sources **100T1** and **100T2** has the upper surface **101** including the light-emitting portion **110**, the lower surface **102** opposite to the upper surface **101**, and the lateral surfaces **103** between the upper surface **101** and the lower surface **102**. For example, the light sources **100T1** and **100T2** have a substantially rectangular parallelepiped shape with four corners curved inward in a plan view. The lateral surfaces **103** of the light sources **100T1** and **100T2** are bonded with the bonding member **60T** disposed therebetween such that the upper surfaces **101** and the lower surfaces **102** of the light sources are exposed from the bonding member **60T** to constitute the single light-emitting device **200T**. In this example, in the light-emitting device **200T**, the lateral surfaces **103**, facing each other, of two adjacent light sources **100T1** and **100T2** having different emission colors are bonded together such that the light-emitting portions **110** are aligned in a single direction.

[0091] By combining light sources having different emission colors, a multicolor light-emitting device **200T** can be obtained. The light-emitting device **200T** can be manufactured by providing a plurality of light sources having emission colors in different chromaticity ranks in the providing step, extracting a plurality of light sources in chromaticity ranks of a desired emission color in the extracting step, and bonding lateral surfaces of the extracted light sources. In this case, the light sources having different emission colors preferably have substantially the same outer shape.

[0092] Each of the light sources **100T1** and **100T2** has a substrate **10T** defining a recess in the upper surface of the substrate **10T**. The light-emitting element **20** and the protective element **25** are disposed in the recess. For example, a ceramic substrate is used for the substrate **10T**. For the substrate **10T** defining the recess, a known substrate such as a ceramic substrate and a resin package can be used.

[0093] Each of the light-emitting element **20** and the protective element **25** is electrically connected to wirings on the substrate **10T** via electroconductive members such as wires.

[0094] A light-transmissive member **30T** covers the light-emitting element **20** and the protective element **25** and transmits light emitted from the light-emitting element **20** to emit the light to the outside. The light-transmissive member **30T** serves as the light-emitting portion of each of the light sources **100T1** and **100T2**. For the light-transmissive member **30T**, a material similar to a material of the light-transmissive member **30** described above. The light-transmissive member **30T** can contain a phosphor adapted to convert wavelengths of at least a portion of light incident on the light-transmissive member **30T**.

[0095] The light-emitting device **200T** includes the two light sources **100T1** and **100T2** having different chromaticities. The light source **100T1** of the light-emitting device **200T** is configured to emit red light. More specifically, as shown in FIG. 5, the light source **100T1** is configured to emit light in a chromaticity range Red defined by a first straight line connecting a first point and a second point, a second straight line connecting the second point and a third point, a third straight line connecting the third point and a fourth point, and a curved line of the chromaticity diagram connecting the fourth point and the first point in the CIE 1931 chromaticity diagram, where the first point has chromaticity coordinates (x,y) of (0.669,0.331), the second point has coordinates (0.652,0.331), the third point has coordinates (0.688,0.296), and the fourth point has coordinates (0.707,0.294). The light source **100T1** can emit red light by using a blue light-emitting element as the light-emitting element **20** and employing the combination of the blue light-emitting element and a phosphor excited by blue light emitted from the light-emitting element to emit red light.

[0096] The light source **100T2** of the light-emitting device **200T** is configured to emit blue-green (turquoise) light. More specifically, as shown in FIG. 5, the light source **100T2** is configured to emit light in a chromaticity range Turquoise defined by a first straight line connecting a first point and a second point, a second straight line connecting the second point and a third point, a third straight line connecting the third point and a fourth point, and a curved line of the chromaticity diagram connecting the fourth point and the first point in the CIE 1931 chromaticity diagram, where the first point has chromaticity coordinates (x,y) of (0.012,0.495), the second point has coordinates (0.200,0.400), the third point has coordinates (0.200,0.320), and the fourth point has coordinates (0.040,0.320). For example, the light source **100T2** can emit blue-green light by using blue-green light-emitting element as the light-emitting element **20**. In order to adjust the chromaticity, a phosphor adapted to be excited by light emitted from the light-emitting element **20** and has a peak emission wavelength within the green, yellow-green, yellow, orange, or red wavelength range may be used in combination with the light source **100T2**.

[0097] The bonding member **60T** is disposed between lateral surfaces of the adjacent light sources **100T1** and **100T2** facing each other and bonds the lateral surfaces **103** of the adjacent light sources **100T1** and **100T2** such that the upper surface **101** and the lower surface **102** of each of the light sources **100T1** and **100T2** are exposed. For the bonding member **60T** used in this example, a member similar to the bonding member **60** described above can be used. In the case in which a light-transmissive member **30T** contains a phosphor, the bonding member **60T** is preferably made of a light-shielding resin. Examples of the light-shielding resin include a black or gray resin containing a black pigment. With such a material, light leaking out from one light source and exciting a phosphor of the other adjacent light source. Accordingly, a light-emitting device with good “visibility,” which refers to good contrast between a light-emitting region and a non-light-emitting region, can be obtained.

[0098] The light-emitting device **200T** having such a structure will be described with reference to FIG. 5 and FIG. 6A to FIG. 6C. In FIG. 6A to FIG. 6C, light sources indicated by hatching with a netlike pattern are being left turned on.

[0099] For example, FIG. 6A to FIG. 6C schematically show a light-emitting unit **300** including a combination of two light-emitting devices **200T**, each shown in FIG. 4A and FIG. 4B, as a first

light-emitting device **200T1** and a second light-emitting device **200T2**. For example, this light-emitting unit **300** can be used for a vehicle lamp for a car or a railway carriage. In this case, the light-emitting unit **300** preferably has a structure in which the light-emitting devices **200T**, each shown in FIG. **4A** and FIG. **4B**, are point-symmetrically arranged at right and left sides to serve as the first light-emitting device **200T1** and the second light-emitting device **200T2**.

[0100] The light-emitting unit **300** emits red light when the first light source **100T1** of each of the first light-emitting device **200T1** and the second light-emitting device **200T2** is turned on. The light-emitting unit **300** emits blue-green light when the second light source **100T2** of each of the first light-emitting device **200T1** and the second light-emitting device **200T2** is turned on. The light-emitting unit **300** can emit white light having a mixed color constituted of red light and blue-green light when the first light sources **100T1** and the second light sources **100T2** of the first light-emitting device **200T1** and the second light-emitting device **200T2** are turned on at the same time.

[0101] For example, chromaticity ranks of light sources to be extracted as the first light source and the second light source are determined such that a straight line passing through a chromaticity point R of the first light source that emits light in the chromaticity range “Red” and a chromaticity point T of the second light source that emits light in the chromaticity range “Turquoise” passes through a chromaticity range “White” of white light in the XY chromaticity diagram as shown in FIG. **5**. For example, the chromaticity range “White” of white light is a range defined by a first straight line connecting a first point and a second point, a second straight line connecting the second point and a third point, a third straight line connecting the third point and a fourth point, and a fourth straight line connecting the fourth point and the first point, where the first point has chromaticity coordinates (x,y) of (0.321,0.306), the second point has coordinates (0.314,0.338), the third point has coordinates (0.346,0.367), and the fourth point has coordinates (0.348,0.341).

[0102] When the light-emitting unit **300** is used for a vehicle lamp, for example, blue-green light can serve as an automated driving indicator lamp indicating to those around the vehicle that the vehicle is in the automatic driving mode, red light can serve as a taillight, and white light can serve as a backup light (reversing light).

[0103] Light-emitting devices employing another combination of colors of light sources may be used for the light-emitting unit. For example, the first light source is configured to emit light having an emission color of orange (amber). More specifically, as shown in FIG. **5**, the first light source can be configured to emit light in a chromaticity range “Amber” defined by a first straight line connecting a first point and a second point, a second straight line connecting the second point and a third point, a third straight line connecting the third point and a fourth point, and a curved line of the chromaticity diagram connecting the fourth point and the first point in the CIE 1931 chromaticity diagram, where the first point has chromaticity coordinates (x,y) of (0.562,0.438), the second point has coordinates (0.549,0.425), the third point has coordinates (0.576,0.407), and the fourth point has coordinates (0.589,0.411). The second light source is configured to emit light having an emission color of light-blue.

[0104] More specifically, as shown in FIG. **5**, the second light source is configured to emit light in a chromaticity range “Light blue” defined by a first straight line connecting a first point and a second point, a second straight line connecting the second point and a third point, a third straight line connecting the third point and a fourth point, and a curved line of the chromaticity diagram connecting the fourth point and the first point, where the first point has chromaticity coordinates (x,y) of (0.050,0.275), the second point has coordinates (0.100,0.290), the third point has coordinates (0.120,0.220), and the fourth point has coordinates (0.069,0.200). In this case, when the first light sources of the first light-emitting device and the second light-emitting device are turned on, the light-emitting unit emits orange light. When the second light sources of the first light-emitting device and the second light-emitting device are turned on, the light-emitting unit emits light-blue light. When the first light sources and the second light sources of the first light-emitting device and the second light-emitting device are turned on at the same time, the light-

emitting unit can emit white light having a mixed color constituted of orange and light-blue. [0105] For example, light-blue light can serve as a lamp that indicates the state of charge of a vehicle, orange light can serve as a turn signal, and white light can serve as a daytime running lamp (DRL).

[0106] As described above, in the light-emitting unit, light sources in such chromaticity ranks that a straight line passing through the chromaticity coordinates of two light sources in each light-emitting device passes through the white chromaticity range in the chromaticity diagram can be used in combination. This structure allows white light and colored light having two colors to be emitted through a single lens of the light-emitting unit, which allows for saving space and improving design qualities.

[0107] In the light-emitting device including a plurality of light sources arranged with bonding members disposed therebetween, influences of turning on and off of adjacent light sources can be reduced, so that brightness and darkness of the individual light sources are clearly observed.

[0108] In the case described above in which a plurality of light sources are used in the light-emitting device, two, three, or four or more light sources are aligned in a straight direction, and adjacent light sources are bonded with bonding members disposed therebetween, but other appropriate arrangement may be employed. For example, light sources may be arranged in two rows and two columns. In particular, when the light sources include light-emitting elements in different luminous flux ranks that can be individually turned on and off, color variations can be increased.

[0109] A method of manufacturing the light-emitting unit is substantially the same as the method of manufacturing the light-emitting device of the first embodiment except that a plurality of light-emitting device collective bodies and light-emitting devices of the same or different emission colors are used, and the description is omitted.

[0110] While certain embodiments of the present invention have been described above, the present invention is not limited the description above, and should be broadly construed on the basis of the claims. The present invention also encompasses variations and modifications that are made on the basis of the description above.

Claims

1. A light-emitting device comprising: a plurality of light sources arrayed in a direction; and a bonding member bonding adjacent ones of the plurality of light sources to each other, wherein each of the light sources has an upper surface including a light-emitting portion, a lower surface opposite to the upper surface, and a lateral surface between the upper surface and the lower surface, wherein each of the light sources comprises an external connection terminal at the lower surface, wherein lateral surfaces of the adjacent ones of the plurality of light sources are bonded via the bonding member such that the upper surfaces and the lower surfaces of the light sources are exposed from the bonding member and such that the bonding member is spaced apart from the external connection terminals, and wherein the upper surfaces of the light sources constitute a part of an upper surface of the light-emitting device.
2. The light-emitting device according to claim 1, wherein each of the light sources comprises: a substrate, a light-emitting element disposed on the substrate, a light-transmissive member covering the light-emitting element, and a covering member from which an upper surface of the light-transmissive member is exposed, the covering member covering lateral surfaces of the light-transmissive member.
3. The light-emitting device according to claim 2, wherein the substrate has an upper surface and a lower surface each comprising a wiring.
4. The light-emitting device according to claim 1, wherein the bonding member has an upper surface constituting a part of the upper surface of the light-emitting device.

5. The light-emitting device according to claim 1, wherein the bonding member has a lower surface constituting a part of a lower surface of the light-emitting device.
6. The light-emitting device according to claim 4, wherein the upper surface of the bonding member is flush with the upper surfaces of the light sources.
7. The light-emitting device according to claim 5, wherein the lower surface of the bonding member is flush with the lower surfaces of the light sources.
8. The light-emitting device according to claim 2, wherein: the covering member has a portion that covers the lateral surface of the light-transmissive member, and a width of the portion in an array direction of the light sources is 30 μm or greater and 300 μm or less.
9. The light-emitting device according to claim 1, wherein a width of the bonding member in an array direction of the light sources is 5 μm or greater and 50 μm or less.
10. The light-emitting device according to claim 1, wherein the light-transmissive member contains a phosphor capable of converting at least a part of wavelengths of light emitted from the light-emitting device.
11. The light-emitting device according to claim 1, wherein each of the light sources comprises a first light source and a second light source having a different chromaticity from the first light source.
12. The light-emitting device according to claim 11, wherein a phosphor contained in a light-transmissive member of the first light source is different from a phosphor contained in a light-transmissive member of the second light source.
13. The light-emitting device according to claim 11, wherein: based on XY chromaticity coordinates of a CIE 1931 chromaticity diagram: chromaticity coordinates of the first light source have a first R point (x,y) of (0.669,0.331), a second R point (x,y) of (0.652,0.331), a third R point (x,y) of (0.688,0.296), and a fourth R point (x,y) of (0.707,0.294), and the first light source emits light within an area enclosed by a first line, a second line, a third line, and a curved line on the CIE 1931 chromaticity diagram, where the first line connects the first R point and the second R point, the second line connects the second R point and the third R point, the third line connects the third R point and the fourth R point, and the curved line connects the fourth R point and the first R point.
14. The light-emitting device according to claim 11, wherein: based on XY chromaticity coordinates of a CIE 1931 chromaticity diagram: chromaticity coordinates of the second light source have a first T point (x,y) of (0.012,0.495), a second T point (x,y) of (0.200,0.400), a third T point (x,y) of (0.200,0.320), and a fourth T point (x,y) of (0.040,0.320), and the second light source emits light within an area enclosed by a first line, a second line, a third line, and a curved line on the CIE 1931 chromaticity diagram, where the first line connects the first T point and the second T point, the second line connects the second T point and the third T point, the third line connects the third T point and the fourth T point, and the curved line connects the fourth T point and the first T point.
15. The light-emitting device according to claim 11, wherein: based on XY chromaticity coordinates of a CIE 1931 chromaticity diagram: chromaticity coordinates of the third light source have a first W point (x,y) of (0.321,0.306), a second W point (x,y) of (0.314,0.338), a third W point (x,y) of (0.346,0.367), and a fourth W point (x,y) of (0.348,0.341), and a combination of the first light source and the second light source emits light within an area enclosed by a first line, a second line, a third line, and a fourth line, where the first line connects the first W point and the second W point, the second line connects the second W point and the third W point, the third line connects the third W point and the fourth W point, and the fourth line connects the fourth W point and the first W point.
16. A vehicle lamp comprising: a light-emitting device comprising: a plurality of light sources; and a bonding member bonding adjacent ones of the plurality of light sources to each other, wherein each of the light sources has an upper surface including a light-emitting portion, a lower surface opposite to the upper surface, and a lateral surface between the upper surface and the lower surface,

wherein each of the light sources comprises an external connection terminal at the lower surface, wherein lateral surfaces of the adjacent ones of the plurality of light sources are bonded via a bonding member such that the upper surfaces and the lower surfaces of the light sources are exposed from the bonding member and such that the bonding member is spaced apart from the external connection terminals, and wherein the upper surfaces of the light sources constitute a part of an upper surface of the light-emitting device.
